Problem Set 1

Section 1.2:-

23) First blank:- $\frac{w_1}{\|w\|}$ Second blank:- $\frac{w_2}{\|w\|}$

$\cos\beta\cos\alpha + \sin\beta\sin\alpha = \cos(\beta-\alpha)$

$\downarrow$

$\frac{w_1}{\|w\|} \times \frac{v_1}{\|v\|} + \frac{w_2}{\|w\|} \times \frac{v_2}{\|v\|}$

$\frac{w_1 v_1}{\|w\| \times \|v\|} + \frac{w_2 v_2}{\|w\| \times \|v\|}$

$\frac{w_1 v_1 + w_2 v_2}{\|w\| \times \|v\|}$

$w_1v_1 + w_2v_2 = v \times w$ by definition of product

$\frac{v \cdot w}{\|w\| \times \|v\|}$

28) Yes it is possible. Let $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $u = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$ and $w = \begin{bmatrix} -1 \\ -5 \end{bmatrix}$.

Then $v \cdot u = (1)(-1) + (0)(5) = -1 < 0$, $u \cdot w = (-1)(-1) + (-5)(5) = -24 < 0$ and $v \cdot w = (1)(-1) + (0)(-5) = -1 < 0$

Section 1.3

4) Let $x_1 = 1$, $x_2 = -2$ and $x_3 = 1$.
Then :-

$$1\begin{bmatrix}1\\2\\3\end{bmatrix} + (-2)\begin{bmatrix}4\\5\\6\end{bmatrix} + 1\begin{bmatrix}7\\8\\9\end{bmatrix}$$

$$= \begin{bmatrix}1\\2\\3\end{bmatrix} + \begin{bmatrix}-8\\-10\\-12\end{bmatrix} + \begin{bmatrix}7\\8\\9\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$$

First choice :- Dependent
First blank :- Plane

13) $$\begin{bmatrix}0 & 1 & 0 & 0 & 0\\ -1 & 0 & 1 & 0 & 0\\ 0 & -1 & 0 & 1 & 0\\ 0 & 0 & -1 & 0 & 1\\ 0 & 0 & 0 & -1 & 0\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\\x_4\\x_5\end{bmatrix} = \begin{bmatrix}b_1\\b_2\\b_3\\b_4\\b_5\end{bmatrix}$$

$x_2 = b_1$
$x_3 - x_1 = b_2$
$x_4 - x_2 = b_3$
$x_5 - x_3 = b_4$
$-x_4 = b_5$

The combination of left sides that gives zero is:-
$x_2 + 0(x_3 - x_1) +$

The combination of left sides that gives zero is:- ~~$x_2 + 0(x_3 - x_1) + (x_4$~~
$x_2 + 0(x_3 - x_1) + (x_4 - x_2) + 0(x_5 - x_3) + (-x_4)$
$x_2 + (x_4 - x_2) + (-x_4) = 0$

Since And so, the combination of right hand sides that gives zero is:-

$b_1 + 0(b_2) + b_3 + 0(b_4) + b_5$
$= b_1 + b_3 + b_5$

Section 2.1

29)

$$u_2 = Au_1 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}$$

$$u_3 = Au_2 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix} = \begin{bmatrix} 0.65 \\ 0.35 \end{bmatrix}$$

The sum of the components of each vector is 1

30)

$$u_4 = Au_3 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.65 \\ 0.35 \end{bmatrix} = \begin{bmatrix} 0.625 \\ 0.375 \end{bmatrix}$$

$$u_5 = Au_4 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.625 \\ 0.375 \end{bmatrix} = \begin{bmatrix} 0.6125 \\ 0.3875 \end{bmatrix}$$

$$u_6 = Au_5 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.6125 \\ 0.3875 \end{bmatrix} = \begin{bmatrix} 0.60625 \\ 0.39375 \end{bmatrix}$$

nu: EVOLVE

$$u_7 = Au_6 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.60625 \\ 0.39375 \end{bmatrix} = \begin{bmatrix} 0.603125 \\ 0.396875 \end{bmatrix}$$

$$v_1 = Av_0 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix}$$

$$v_2 = Av_{x1} = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix} = \begin{bmatrix} 0.45 \\ 0.55 \end{bmatrix}$$

$$v_3 = Av_2 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.45 \\ 0.55 \end{bmatrix} = \begin{bmatrix} 0.525 \\ 0.475 \end{bmatrix}$$

$$v_4 = Av_3 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.525 \\ 0.475 \end{bmatrix} = \begin{bmatrix} 0.5625 \\ 0.4375 \end{bmatrix}$$

$$v_5 = Av_4 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.5625 \\ 0.4375 \end{bmatrix} = \begin{bmatrix} 0.58125 \\ 0.41875 \end{bmatrix}$$

$$v_6 = Av_5 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.58125 \\ 0.41875 \end{bmatrix} = \begin{bmatrix} 0.590625 \\ 0.409375 \end{bmatrix}$$

$$v_7 = Av_6 = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.590625 \\ 0.409375 \end{bmatrix} = \begin{bmatrix} 0.5953125 \\ 0.4046875 \end{bmatrix}$$

From the MATLAB graphs, the steady state vector appears to be $(0.6, 0.4) = s$.

$$As = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

Hence confirmed

Section 2.2

20) First blank:- Linear combination

$2x-4y-\cancel{4}2z=4$ (LHS is 2x the LHS of $x-2y-z=1$ but RHS is 4x)

32) a) Zero b) Zero ~~d~~

c) Construct a 100x100 matrix with the first colum having any non zero entries, then make the second colum the same but negative. ~~Make the~~ Fill in the rest of the entries with any non zero values

~~d~~ c) Construct a 100x100 matrix such that every column is the vector $(1, 2, 3 \ldots 99, 100)$

d) Row picture:- A 99D hyperplane through the origin in R^{100}

Column picture:- 100 ~~vecto~~ identical vectors all pointing in the same direction.

Section 2.3

22) a) $\sum_{j=1}^{n} a_{3j} x_j$

b) $a_{21} - 2a_{11}$

c) $a_{21} - 2a_{11}$

d) $\sum_{j=1}^{n} a_{1j} x_j$

29) $E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$

Name the previous E 'E_1'. Then:-

$$E_2(E_1P) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$E_3(E_2E_1P) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$M = E_3 \times E_2 \times E_1$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$

Section 2.4

32) $Ax = I$ (the identity matrix)

36) a) $(AB)C$ has $mnprmpq$ multiplications and $A(BC)$ has a multiplication count of $mnq + npq$ which evaluates to 196 and 360 respectively and so $(AB)C$ is preferred

b) The multiplication count for $(u^T v)w^T$ is $2N$ while the multiplication count for $u^T(vw^T)$ is $2N^2$. Therefore we prefer $u^T(vw^T)$

c) $\frac{\text{Cost of } (AB)C}{mnpq}$ $\frac{\text{Cost of } A(BC)}{mnpq}$

$\frac{mnp + mpq}{mnpq}$ $\frac{mnq + npq}{mnpq}$

$\frac{mnp}{mnpq} + \frac{mpq}{mnpq}$ $\frac{mnq}{mnpq} + \frac{npq}{mnpq}$

$\frac{1}{q} + \frac{1}{n}$ $\frac{1}{p} + \frac{1}{m}$

$q^{-1} + n^{-1}$ $p^{-1} + m^{-1}$

$(AB)C$ is faster when cost of $(AB)C$ is less than cost of $A(BC)$ i.e. when

$$mnp + mpq < mnq + npq$$

$$\frac{mnp + mpq}{mnpq} < \frac{mnq + npq}{mhpq}$$

$$q^{-1} + n^{-1} < p^{-1} + m^{-1}$$

(As shown by the above algebra)

Section 2.5

a)

7) ~~Suppos~~ Let A be any matrix with row 3 = row 1 + row 2 and ~~$Ax = (0,0,$~~ $Ax = (1,0,0)$. Suppose $Ax = (1,0,0)$ has a solution. ~~Then~~ i.e. 'x'

~~Then~~ Let the rows of A be denoted A_1, A_2 and A_3. Then $A_1x = 1$, $A_2x = 0$, and $A_3x = 0$. Since $A_1x = 1$ and $A_2x = 0$, $A_1x + A_2x = 1 + 0 = 1$. And since A_3x = $A_1 + A_2$, $A_3x = A_1x + A_2x = 1$, but that contradicts the fact that $A_3x = 0$ and hence our supposition is false i.e. $Ax = (1,0,0)$ has no solution.

b) $b = (1,0,1)$ would allow a solution. (or more generally, whenever $b_3 = b_1 + b_2$)

c) We get a 0 in the pivot position of row 3 (i.e. in the eliminated matrix), row 3 is $(0,0,0)$